

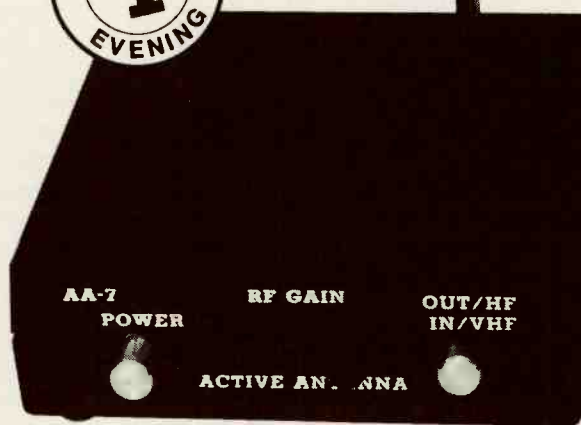
If you have a shortwave or high-frequency receiver or scanner that is struggling to capture signals with a short, whip antenna, and you'd like the kind of performance that a 60-foot long-wire antenna can provide but lack the space to put one up, consider building the AA-7 HF/VHF/UHF Active Antenna described in this article.

The AA-7 is a relatively simple antenna that is designed to amplify signals from 3 to 3000 megahertz (MHz), including three recognized ranges: 3–30 MHz high-frequency (HF) signals; 30–300 MHz very-high frequency (VHF) signals; 300–3000 MHz ultra-high frequency (UHF) signals. Those bands are typically occupied by shortwave, ham, government, and commercial radio signals.

Active Antennas. In its simplest form, an active antenna uses a small whip antenna that feeds incoming RF to a preamplifier, whose output is then connected to the antenna input of a receiver. Unless specifically designed otherwise, all active antennas are intended for receive-only operation, and thus should not be used with transceivers; transmitting into an active antenna will probably destroy its active components.

A well-designed broadband active antenna considers field strength of the desired signal (measured in microvolts per meter of antenna length), atmospheric and other noise, diameter of the antenna, radiation resistance, and antenna reactance at various frequencies, plus the efficiency and noise figure of the amplifier circuit itself.

Circuit Description. Figure 1 shows a schematic diagram of the AA-7, which contains only two active elements; Q1 (an MFE201 N-channel dual-gate MOSFET) and Q2 (a 2SC2570 NPN VHF silicon transistor). Those transistors provide the basis of two independent, switchable RF preamplifiers. Two double-pole double-throw (DPDT) switches play a major



ACTIVE ANTENNA

*Lift hard-to-hear signals
out of the mud with
this handy receiver accessory.*

BY FRED BLECHMAN

role in the operation of the AA-7. Switch S1 is used to select one of the two preamplifier circuits (either HF or VHF/UHF). Switch S2 is used to turn off the power to the circuit, while coupling the incoming RF directly to the input of the receiver. That gives the receiver non-amplified access to the auxiliary antenna jack, at J1, as well as the on-board telescoping whip antenna.

With switch S2 in its power-on position, the input and output jacks are disconnected and B1 (a 9-volt transistor-radio battery) is connected to the circuit. With switch S1 in the position shown in the schematic, incoming RF is directed to the HF preamp circuit built around Q1 (an MFE201 N-channel dual-gate MOSFET). The HF preamp operates with an exceptionally low noise level, and is ideal for copying weak CW and single-sideband signals.

When S1 is switched to the other position, the captured signal is cou-

pled to the VHF/UHF preamp built around Q2 (a 2SC2570 NPN VHF silicon transistor), which has excellent VHF through microwave characteristics. With the on-board whip antenna adjustable to resonance through much of the VHF-UHF region (length in feet = 234 divided by the frequency in MHz), the VHF/UHF mode is ideal for indoor and portable use with VHF scanners and other receivers.

Either mode can be used when tuning 3–30-MHz HF signals. The VHF/UHF preamp offers higher gain than the HF preamp, but also has a higher noise level. You can easily choose either amplifier for copying any signal of interest—just try both S1 positions. The RF gain control (R5) can be used to trim the output of either amplifier.

Caution: The AA-7 is not intended for transmitting operations (be it ham, maritime, or CB); if it is used with a transceiver of any kind, make sure it is not possible to transmit by accidentally pressing a mike button or CW keyer. Transmitting RF into the AA-7 is likely to ruin one or both of the transistors in the circuit.

Construction. The AA-7, which can be built from scratch or purchased in kit form from the supplier listed in the Parts List, was assembled on a printed-circuit board, measuring about 4 by 4 1/8 inches. A template for the printed-circuit board is shown in Fig. 2. You can either etch your own board from that template, or purchase the circuit board or a complete kit of parts (which includes a printed-circuit board and all parts).

The kit comes with a 16-page kit assembly and instruction manual that gives step-by-step assembly instructions and contains additional information not covered in this article. Kit assembly time, working slowly and carefully, should take less than an hour.

Most of the parts specified in the Parts List are standard components and can be procured through conventional hobby electronics suppliers.

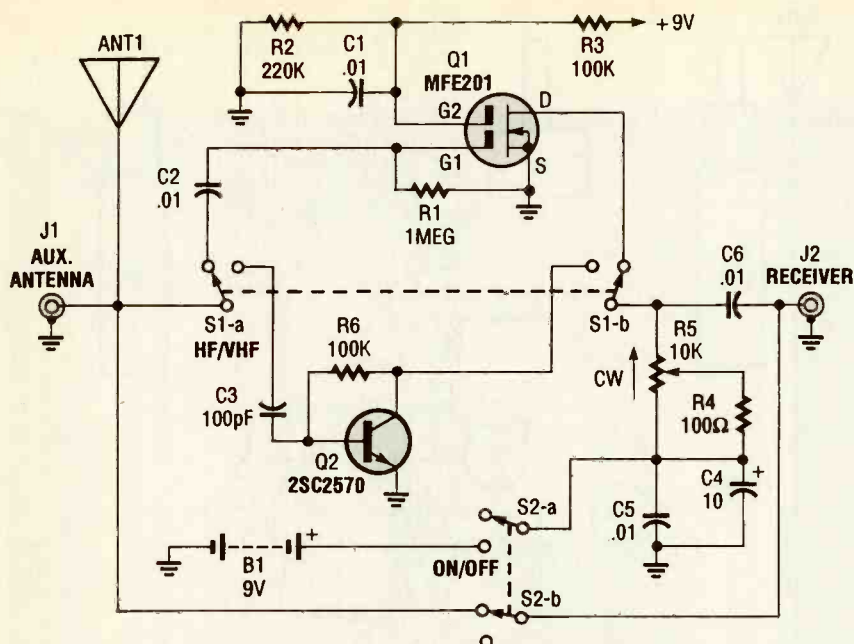


Fig. 1. The AA-7 Active Antenna contains only two active elements; Q1 (an MFE201 N-channel dual-gate FET) and Q2 (a 2SC2570 NPN VHF silicon transistor), which provide the basis of two independent, switchable RF preamplifiers.

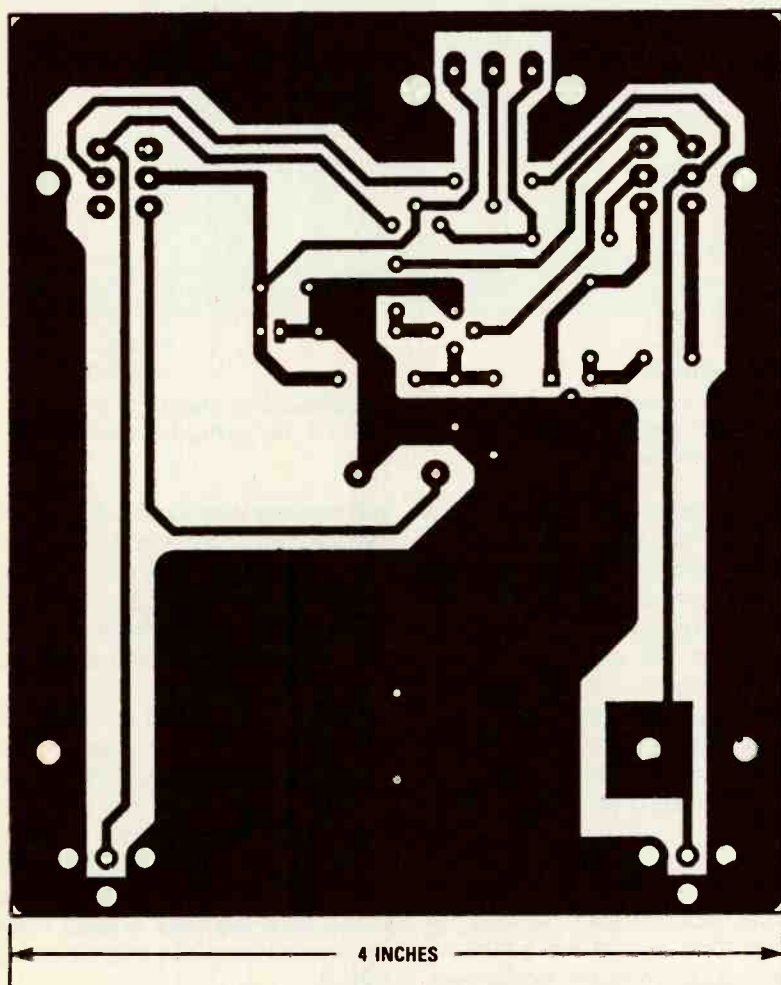


Fig. 2. The AA-7 was assembled on a printed-circuit board, measuring about 4 by 4 1/16 inches. A template for the printed-circuit board is shown here.

PARTS LIST FOR THE AA-7 ACTIVE ANTENNA

SEMICONDUCTORS

Q1—MFE201, 3N204, or SK3991 N-channel, dual-gate MOSFET (see text)*

Q2—2SC2570, 2N5179, or SK9139 NPN VHF silicon transistor (see text)*

RESISTORS

(All fixed resistors are 1/4-watt, 5% carbon units.)

R1—1-megohm

R2—220,000-ohm

R3, R6—100,000-ohm

R4—100-ohm

R5—10,000-ohm potentiometer*

CAPACITORS

C1, C2, C5, C6—0.01-μF, ceramic-disc

C3—100-pF ceramic-disc

C4—4.7- to 10-μF, 16-WVDC, radial-lead electrolytic

ADDITIONAL PARTS AND MATERIALS

MATERIALS

B1—9-volt transistor-radio battery

S1, S2—DPDT PC-mount pushbutton switch*

J1, J2—PC-mount RCA jack*

ANT1—Telescoping whip antenna (screw mount)*

Printed-circuit materials, enclosure, battery holder and connector, wire, solder, hardware, etc.

Note: The following items are available from Ramsey Electronics, Inc. (793 Canning Parkway, Victor, NY 14564; Tel. 716-924-4560): A complete kit of parts (order #AA-7BP), including printed-circuit board, \$24.95; etched and drilled printed-circuit board only (order #AA-7PCBP), \$10.00; special parts kit, containing all items marked with an asterisk (*) in the Parts List (order #AA-7SPKBP), \$14.50; custom case and knob set (order #CAA-BP), \$12.95. Please add \$3 for orders under \$20, and \$3.75 for postage/handling. New York residents, please add appropriate sales tax.

However, some parts—J1, J2, S1, S2, and R5—have particular physical mounting dimensions; the board is designed to accept those parts. In addition, Q1 and Q2 can be hard to find; however, it is possible to make substitutions provided that you can find a supplier. Suitable replacements

for Q1 and Q2 are given in the Parts List.

The telescoping whip antenna screw-mounts to the board; the screw provides contact between the printed-circuit board traces and the antenna. To save time and trouble locating and ordering hard-to-find parts, a Special Parts Kit is also offered by the supplier listed in the Parts List.

A parts-placement diagram for the AA-7's printed-circuit board is shown in Fig. 3. When assembling the circuit, be especially careful that transistors Q1 and Q2, and electrolytic capacitor C4, are oriented as shown.

Although not shown in the schematic (Fig. 1) or the parts-placement (Fig. 3) diagrams, an optional LED power indicator can be added to the circuit. Adding a power indicator to the circuit allows you to tell at a glance if the circuit is on; leaving the circuit on, even though the AA-7 draws only about 0.7 mA, will eventually discharge the battery. Of course, adding an LED will increase the current drain by about 7 mA, but the red glow makes it obvious when the unit is on.

If you decide to include the indicator in your project, power for the indicator can be easily taken from the switched 9-volt DC terminal of S2 (center terminal, right side, looking at the top of S2). Simply connect the positive voltage to the anode (longer wire) of the LED and connect the cathode lead through a current-limiting resistor of about 1000 ohms to a ground point on the printed-circuit board, or as the author did from the frame of R5. Mount the LED at any

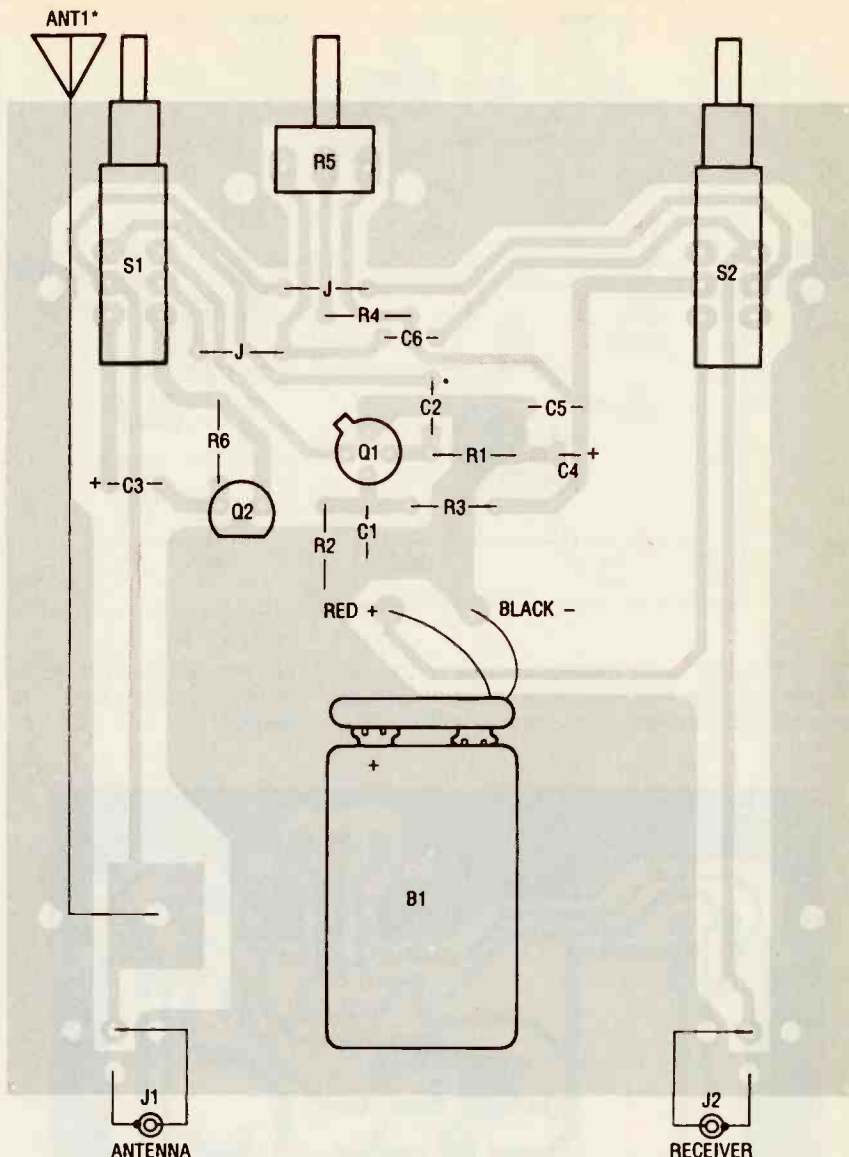


Fig. 3. When assembling the AA-7's printed-circuit board, be especially careful that transistors Q1, and Q2, and electrolytic capacitor C4, are oriented as shown in this parts-placement diagram.

convenient point near the switch.

Although not supplied with the kit, a custom plastic enclosure (with front and back panels, and knobs for the switches and gain control) is offered in the Parts List. The enclosure comes pre-drilled and silk-screened with the appropriate legends for all the circuit controls and connectors, but is not equipped with holes for the whip antenna or the LED (if you include one).

Use. Prepare a coaxial cable to connect the RF output of the AA-7 to the antenna input of your receiver or scanner. One end of the interconnecting cable must be terminated with an RCA phono plug; the other end connector depends on the tar-

get receiver. With some receivers, the only practical connection is to clip the output of the AA-7 to the receiver's antenna, although that connection method won't be as effective as conventional (ground-return type) coupling.

To increase signal strength, especially for the lower frequencies, you can connect a simple supplementary portable antenna of any design (a dipole, a random-length wire with Earth ground, a bigger vertical whip of some kind, etc.) to the circuit. Just use a small-diameter coaxial cable terminated in an RCA plug for mating with J1.

No alignment is required. If you're
(Continued on page 112)

TYPES OF ACTIVE ANTENNAS

The RF preamplifier section of an active antenna may be of several different styles. A fixed-tuned preamplifier has the greatest gain for the specific frequency of operation, but is limited to only that frequency. A variable-tuned amplifier can be peaked within a range of frequencies, but the circuitry is more complex and expensive, and tuning requires constant resetting whenever the receiver frequency is changed. A broadband preamplifier doesn't require any tuning, making it very versatile and easy to use, but has less gain at any specific frequency. The AA-7 described here, for simplicity and low cost, is of a dual broadband design.

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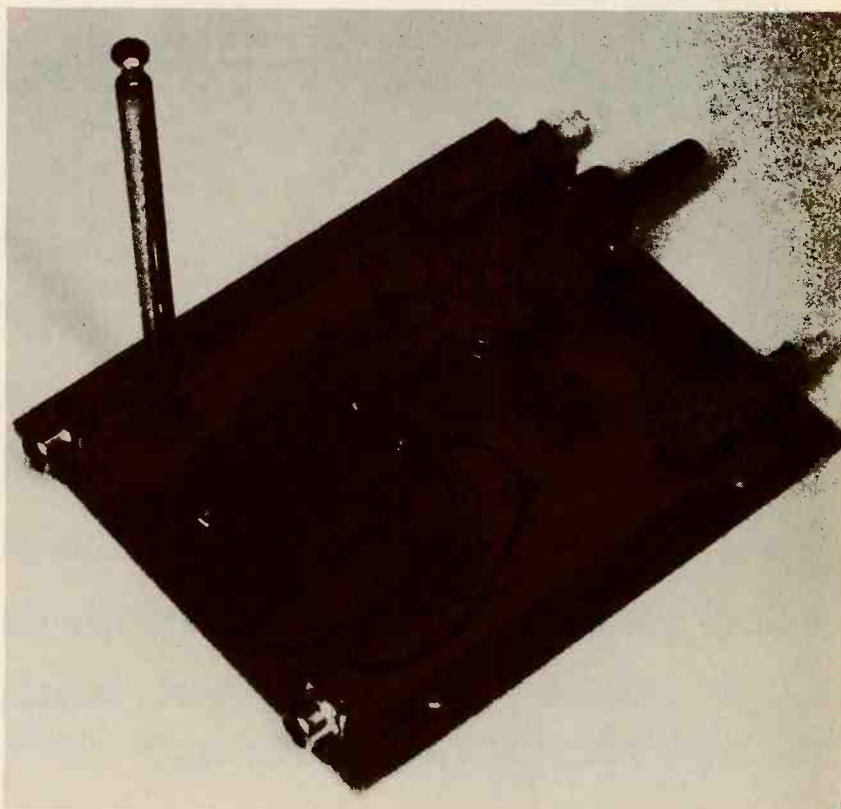
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BUILD AN ACTIVE ANTENNA

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using the whip antenna, simply connect the output of the AA-7 to your receiver, with the unit turned off (that's the bypass position) and the RF gain control (R5) turned fully counter-clockwise. Turn on the receiver and tune-in a weak station. Switch S2 on, and adjust the gain control clockwise to increase the output signal. Toggle S1 back and forth to see which setting gives you the best results. Don't be surprised if the gain control overloads the receiver; if so, back it off.

There are other characteristics or phenomena associated with preamplifiers and active antennas that does not mean that your the circuit is malfunctioning. For example, if you have strong AC hum in the HF setting, the antenna is too close to an AC cord or power line. HF signals may be clearer at the VHF/UHF setting than at the HF setting. Why? Although either preamp, may be used for HF, the signal strength will be greater with the VHF/UHF preamp. However, the HF signal-to-noise ratio is better with the dual-gate-MOSFET-based preamp. Try both and use the best for your particular receiving conditions.



Although not shown in the schematic or the parts-placement diagrams, an LED can be added to the circuit so that you can tell at a glance if the circuit is on. In his own unit, the author accomplished that by connecting the LED (with a series-connected current-limiting resistor) between a terminal on the power switch and the grounded frame of the gain control (R5).

Troubleshooting. The fact that there are two independent preamplifiers in the AA-7 makes faults easier to diagnose than with many other devices. If a problem occurs, only at one setting of S1, concentrate on that part of the circuit. If the problem is common to both settings, the components and connections common to both preamps should be checked. Make sure the jumper wires are in place.

Some portable receivers not enclosed in metal cases may break into oscillation when connected to any RF preamplifier. Try reducing the AA-7's gain and make sure that good grounds are provided with the interconnecting coax cables. A preamplifier will intensify any problems due to poor receiver design: overloading, images, or any problems with selectivity and image rejection. ■